

NILAVA METYA

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INTRODUCTION

I identify as an applied mathematician. My research encompasses applying techniques from algebra and geometry to algorithmic and statistical problems in *optimization* and *machine learning* from a structural perspective. More specifically, I like to borrow tools from algebra, convex geometry, Banach space theory and probability.

EDUCATION

Rutgers, the State University of New Jersey - New Brunswick

Doctor of Philosophy in Mathematics

(passed written qualifying exams in first attempt just before program started)

Sep '22 – (expected) '27

Piscataway, New Jersey, USA

Chennai Mathematical Institute

Bachelor of Science (Honours) in Mathematics and Computer Science | CGPA: 9.72/10

Position: *Third* (out of 55 students)

(completed degree requirements in 2.5 years)

Aug '19 – May '22

Chennai, Tamil Nadu, India

Don Bosco School, Liluah

Indian School Certificate (ISC) 2019 | *Percentage*: 97.25%

Position: *First* in science stream (~ 55 students), *second* overall (~ 180 students)

Indian Certificate of Secondary Education (ICSE) 2017 | *Percentage*: 96.6%

Position: *First* in school (~ 180 students)

Apr '06 – Mar '19

Howrah, West Bengal, India

CURRENT AND UPCOMING RESEARCH

9. *N Metya, S Mukherjee. Optimization to fairness for a convolution of two dice. 2026+. *Upcoming*
8. *N Metya, A Sengupta. Improved constructions for higher order MDS codes. 2026+. *Upcoming*
7. N Metya, O Leong, Y S Soh. Convex optimization over the space of convex bodies. 2026+. *Upcoming*
6. N Metya. ML degrees via chordal completion. 2026+. *Upcoming*
5. *N Metya, J Saunderson. A new Hyperbolic Chernoff bound. 2026+. *Upcoming*
4. *S Gaungly, N Metya, A Morozov, A Sengupta. Activation Functions Control Finite-Width Concentration in Wide Neural Networks. 2026.
4th Workshop on High-dimensional Learning Dynamics at ICML 2026
3. M Smart, S Gaungly, N Metya, A Morozov, A Sengupta. Attention as In-Context Empirical Bayes: A Two-Stage View via Particle Dynamics. 2026.
ICML 2026 Workshop on Foundations of Deep Generative Models
2. N Metya, A Sinha. Temporal robustness in discrete time linear dynamical systems. 2025.
Submitted
1. *G DePaul, S Hoşten, N Metya, I Nometa. Degrees of the Wasserstein distance to small toric models. 2024.
Journal of Algebraic Statistics

*Alphabetical.

TALKS

INFORMS invited session on AI for Security and Resilience in Complex Networks <i>Temporal robustness in discrete time linear dynamical systems.</i>	Oct '25
ACMU, Indian Statistical Institute Host: Sourav Chakraborty <i>Temporal robustness in discrete time linear dynamical systems.</i>	Aug '25
School of Computing, National University of Singapore Host: Jonathan Scarlett <i>Temporal robustness in discrete time linear dynamical systems.</i>	May '25
Mina Rees Women and Math Conference, CUNY <i>Degrees of the Wasserstein distance to small toric models.</i>	Mar '25

ACADEMIC VISITS

School of Computing, National University of Singapore <i>Prof Jonathan Scarlett</i>	May-Aug '26
Department of Mathematics, National University of Singapore <i>Prof Yong Sheng Soh</i>	May-Aug '26
School of Computing, National University of Singapore <i>Prof Jonathan Scarlett</i>	May-Aug '25
Department of Mathematics, Sorbonne University <i>Prof Anne Marie Aubert</i>	May '22

ATTENDANCE IN CONFERENCES/WORKSHOPS

Annual Summer School on Mathematical Aspects of Data Science <i>Summer School</i> NUS, Singapore	Jul '26
Princeton Machine Learning Theory Summer School <i>Summer School</i> Princeton	Aug '24
Efficient Algorithms for High Dimensional Metrics <i>Workshop</i> DIMACS, Rutgers, New Brunswick	May '24
Bayesian Statistics and Statistical Learning <i>Workshop</i> IMSI, Chicago	Dec '23
Algebraic Statistics for Ecological and Biological Systems <i>Workshop</i> IMSI, Chicago	Oct '23
Apprenticeship Week: Varieties from Statistics IMSI, Chicago	Oct '23
Invitation to Algebraic Statistics and Applications IMSI, Chicago	Sep '23
Algebraic Methods in Biochemical Reaction Networks MPI, Leipzig	Jun '23
Joint Mathematics Meetings Boston	Jan '23
AlGeCom-XII (Algebra Geometry and Combinatorics day) UIUC	Oct '22

SELECTED COURSEWORK

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- | | | |
|---------------------------|---|---|
| • Quantum Computation | • Haskell | • Topological Data Analysis ^{Rutgers} |
| • Matrix Computations | • Algorithm Design + Analysis | • Convex Optimization ^{Princeton} |
| • Statistics with R | • Algebraic Geometry ^{Princeton} | • Convexities in Banach Spaces ^{Princeton} |
| • Algebraic Topology | • Homological Algebra ^{Rutgers} | • Random Matrices ^{Princeton} |
| • Complex Analysis | • Probability ^{Rutgers} | |
| • Quiver Representations | • Data Mining ^{Rutgers} | |
| • Algebraic Number Theory | | |

TEACHING AND GRADING

Workshop leader for Calculus II Rutgers		Sep – Dec ’23, Jan – Apr ’24	
Instructor – PhD written qualifying exam in Algebra Rutgers University		Jun – Aug ’23	
Head Counselor at PROMYS India IISc Bangalore		May – Jun ’23	
Counselor at PROMYS Boston University		Jul – Aug ’20, ’21	
Grader Rutgers University			
Information Theoretic methods in Complexity Theory		Sep – Dec ’26	
Graduate Algebra I		Sep – Dec ’25	
Linear Algebra and Applications		Jan – Apr ’25	
Mathematical Statistics		Sep – Dec ’24	
Algebra II		Jan – Apr ’24	
Linear Algebra and Applications		Sep – Dec ’23	
Analysis II		Jan – Apr ’23	
Topics in Applied Algebra		Jan – Apr ’23	
Topology		Sep – Dec ’22	
Theory of Numbers		Sep – Dec ’22	
Teaching Assistant Chennai Mathematical Institute			
Algebra II (Group theory)	BSc 1st year	Prof Manoj Kummini	Jan – May ’22
Algebra I (Linear algebra)	BSc 1st year - head tutor	Prof T R Ramadas	Sep – Dec ’21
Functional Programming in Haskell	BSc and MSc Comp. Sci. 1st year	Prof S P Suresh	Sep – Dec ’21
Probability Theory	BSc 1st year	Prof P Sankaran	Apr – Jul ’21
Discrete Mathematics	BSc 1st year	Prof K V Subrahmanyam	Apr – Jul ’21
Design and Analysis of Algorithms	MSc Data Science 1st year	Prof G Philip	Apr – Jul ’21
Algebra I (Linear algebra)	BSc 1st year	Prof T R Ramadas	Dec ’20 – Mar ’21
Functional Programming in Haskell	BSc and MSc Comp. Sci. 1st year	Prof S P Suresh	Dec ’20 – Mar ’21

HONOURS AND AWARDS

Nominated by Rutgers Math department for SLMATH summer school	Jun '23
<i>Summer school at Leipzig - awarded full travel funding. Only two students from Rutgers Math were fully funded by SLMATH.</i>	
Academic Excellence Award at Rutgers	Sep '22
<i>Received a certificate and \$100 based on performance in Written Qualifying Exams.</i>	
Shriram Scholarship at CMI	'19 – '22
<i>Received institutional fee waiver and monthly stipend (based on entrance exam).</i>	
Ranked 4th nationally at the Bachelor of Statistics (B.Stat.) entrance examination	'19
<i>Indian Statistical Institute (ISI)</i>	
Informatics Olympiad	'17, '18, '19
<i>Selected among (approx) top 100-130 school students in India in Zonal Informatics Olympiad (ZIO).</i>	
Mathematical Olympiad	Jan '18
<i>Selected for Indian National Mathematical Olympiad (INMO) Training Camp top 30 school students in West Bengal.</i>	
Program in Mathematics for Young Scientists (PROMYS)	'18, '19, '20, '21
<i>Awarded the Tara and Jasubhai Mehta Fellowship to PROMYS (among 5 Indian school students in 2018) based on a competitive process. Participated twice as a student ('18, '19) and twice as a counselor ('20, '21).</i>	

Others

- Qualified for **International Collegiate Programming Contest (ICPC)** Kharagpur regionals and Amritapuri regionals in 2019 and secured rank 35 among (approx) 90 university teams at Kharagpur.
- Selected among top 30 students in India to participate in **Scholastic Test of Excellence in Mathematical Sciences (STEMS)** camp at CMI in 2018, based on a competitive exam (across grades 9 – 12 and across Math, Physics, Computer Science).
- Secured the **third position** in **Mathematics Talent Reward Programme (MTRP)** 2016, organized by ISI Kolkata, based on a competitive exam and quizzes at a camp.

SERVICE

Coffee Hour (departmental socializing)	<i>Coordinator and Volunteer</i> Rutgers Math Department	Jan '25 -
Physics of Learning Theory (PhyLT)	<i>Organizer</i> Rutgers Math Department	Jan '25 -
Algebra 'N' Geometry Learning Seminar (ANGeLS)	<i>Organizer</i> Rutgers Math Department	Jan - Apr '23
Student Seminar	<i>Organizer</i> Chennai Mathematical Institute	Oct - Dec '22
ICO Camp (online)	<i>Combinatorics teacher</i> CodeChef	Nov '20

SKILLS

Languages	Bengali (mother tongue), English (fluent), Hindi (fluent)
Programming	JAVA, C++, Python, Haskell, R, HTML, SageMath, Macaulay2, MATLAB
Documentation	$\text{\LaTeX}$, Microsoft Word